# Ali Tleis

Boston, MA
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**Security Clearance:** DoD Investigation in Progress

## Education

**Northeastern University** — Boston, MA
*Bachelor of Science in Computer Science and Sociology* — *Expected Aug. 2028*
- **Coursework:** Algorithms & Data Structures, Object-Oriented Design, Artificial Intelligence, Discrete Structures

## Experience

**MIT Lincoln Laboratory** — Lexington, MA
*Web Application Developer (AI Integration) Co-op* — *Jun. 2026 – Present*
- Shipped internal AI search used by **4,500+ engineers**, owning the technical proposal, implementation, and production release of Java/Spring Boot APIs and Python/FastAPI services.
- Cut query latency **70%** and model token consumption **40%** by combining hybrid Solr keyword/vector retrieval with Sentence Transformers reranking in Python/FastAPI.
- Preserved document access controls in search results and generated answers with permission-aware retrieval tied to existing identity services and incremental Norconex ingestion.
- Deployed Docker containers to existing Kubernetes infrastructure with Helm and GitLab CI/CD, adding automated integration tests and Prometheus/Grafana dashboards to support rollout and diagnose failures.
- Integrated internally hosted Llama inference through vLLM with streamed responses, retrieval-grounded prompts, and Redis caching for repeated queries.

**Top Choice Realty** — Staten Island, NY
*Frontend Developer Intern* — *Jun. 2025 – Sep. 2025*
- Reduced manual reconciliation time **30%** by automating MongoDB synchronization with Python and C# Azure Functions and refactoring schemas to prevent duplicate client and agent records.
- Implemented role-based access and document uploads in Express APIs backed by MongoDB Atlas, storing uploaded files in Azure Blob Storage.
- Added GitHub Actions workflows for automated builds, API tests, and deployment checks across the application.
- Delivered Next.js client-intake and agent-dashboard workflows using reusable React/TypeScript components, standardizing UI behavior and incorporating agent feedback into production releases.

**Robert DeFalco Realty** — Staten Island, NY
*Computer Technician Intern* — *Jun. 2023 – Sep. 2023*
- Automated provisioning for **20+ systems** using PowerShell, standardizing software installation, account setup, and Windows/Linux workstation configuration.
- Built a Python inventory utility to flag missing software, outdated systems, and configuration differences while supporting Microsoft Intune enrollment and Active Directory account administration.

## Projects

**EternalMonitor** | eternalmonitor.dev | GitHub — Open-source contributor
- Owned AMD GPU support end to end for an app with **500+ users** and **100+ GitHub stars**, implementing Rust/FFmpeg AMF H.264/HEVC encoding, keyframe recovery, and software fallback.
- Validated AMD streaming on physical Windows and iPad hardware across DXGI capture, GPU encoding, Wi-Fi/USB transport, VideoToolbox decoding, and Metal rendering.

**Eternal2x** | eternal2x.com — Jan. 2026 – Present
- Automated **2x** upscaling and timeline regrouping in DaVinci Resolve with Python, Lua, OpenCV, and FFmpeg, applying Optical Flow to motion segments and Nearest interpolation to static frames.

## Technical Skills

**Languages:** Java, Python, Rust, C++, C#, TypeScript, JavaScript, SQL, PowerShell, Lua
**Frameworks:** Spring Boot, FastAPI, Node.js/Express, React, Next.js
**Databases & Search:** Apache Solr, Redis, MongoDB Atlas, PostgreSQL
**AI:** Sentence Transformers, vLLM, Llama, Gemini API
**Infrastructure:** Azure Functions, Azure Blob Storage, Docker, Kubernetes, Helm, Linux
**Tools:** GitLab CI/CD, GitHub Actions, Prometheus, Grafana